



VaultofCodes

Assignment 1:

Task1:

Review the following codes, find and fix errors and explain the error

Code Snippet 1: Variable Name Typo

Code:

```
python

number_of_apples = 5
print(number_of_apple)
```

Error to Expect: NameError

Hint: Ensure that variable names are spelled consistently throughout your code. Did you reference the variable correctly when trying to print it?

Code Snippet 2: Accessing List Elements Out of Range

Code:

```
python

fruits = ["apple", "banana", "cherry"]
print(fruits[3])
```

Error to Expect: IndexError

Hint: List indices start at 0. If you're trying to access the third element, what index should you use? Also, consider how many elements are actually in the list.

Debugging Exercise 3: Function Not Behaving as Expected

```
python

def find_average(numbers):
    sum = 0
    for number in numbers:
        sum += number
    average = sum / len(numbers)
    return average

numbers = [1, 2, 3, 4, 5, "6"]
average = find_average(numbers)
print(f"The average is: {average}")
```

Error to Expect: TypeError

Hint: This code attempts to calculate the average of a list of numbers. However, one of the elements in the list is not a number in the conventional sense. How does Python treat different data types when performing arithmetic operations? Consider how you can ensure all elements are appropriately handled for the calculation.

Exercise 4: Incorrect Dictionary Usage

Code:

```
python

def update_record(records, name, score):
    if name in records:
        records[name].append(score)
    else:
        records[name] = score

student_records = {"Alice": [88, 92], "Bob": [70, 85]}
update_record(student_records, "Charlie", 91)
update_record(student_records, "Alice", 95)

print(student_records)
```

Error to Expect: AttributeError

Hint: This function is supposed to update a dictionary containing student names as keys and a list of scores as their values. When adding a score for a student not previously in the records, the code does not correctly handle the assignment. How should you structure the value associated with each key in the dictionary, especially when introducing a new key?

Write the corrected code in a file with explanations of errors

Format:

Write the error

Write corrected code

Explain the error

Submit this file in pdf or Word.

Task2:

Create a presentation on topic

-Introduction and history of Python

-Functions & Modules

Submit in ppt/pdf format